

Who Speaks or How They Speak? Residual Pitch Variability and Electoral Outcome in U.S. Presidential Debates, 1988–2020

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Abstract—Do linguistic and vocal properties distinguish candidates whose tickets went on to win a U.S. presidential election from those whose tickets did not? We study every televised U.S. general-election presidential debate from 1988 to 2020: 24 debates, 29,672 time-aligned sentences, of which 24,193 are spoken by candidates. We screen 43 linguistic, semantic, discourse and acoustic measures for a within-debate difference between winning and non-winning candidates, and then characterise, at the level of the pitch contour itself, the one property that survives. We hypothesise that generalisation to an unseen speaker or debate cannot be assumed and must be tested directly, so we evaluate every predictive model under two holdout schemes: holding out a debate, and holding out every debate containing either of its speakers. We apply the same test to a large language model asked to judge the same debates, together with a contamination probe and a closed-vocabulary grounding check on its stated reasons.

We find that winning candidates carry less residual pitch movement than non-winning candidates once each sentence’s own rise or fall is removed. Of 53 properties tested, this is the only one that survives every validity and stability check we run, including dropping any one of the corpus’s fourteen speakers in turn. It does not, however, predict a debate we have not seen: its 95% prediction interval includes zero, because speaker identity alone explains 81% of the outcome label in a corpus where only fourteen people have ever taken this stage. We demonstrate that the same limitation defeats a decision forest, a linear discriminant and a weighted sum, all three of which call debates correctly until every debate containing either candidate’s own voice is removed from training, at which point all three fall to chance. A large language model given an anonymised transcript calls the winner in 24 of 24 debates; asked instead, with no prediction question attached, to name the debate itself, it also succeeds in 24 of 24 cases, and given only numeric descriptors with nothing to identify, its accuracy falls back to chance. Thus apparent predictive success in this corpus is, three times over, an artefact of speaker or transcript identity leaking into what was meant to be a held-out fold, and the residual-pitch finding itself should be read as a robust within-corpus association rather than a working predictor.

Index Terms—Political discourse analysis, spoken corpus construction, prosody, pitch range, dialogue-act annotation, multi-modal analysis, speaker confound, measurement validity.

I. INTRODUCTION

A broad multimodal screen of U.S. presidential debate speech turns up one coherent, outcome-associated pattern:

The author is a Ph.D. researcher in computational linguistics working on the discourse analysis of U.S. political debates. Corpus, annotation layers and analysis code are deposited as described in Appendix I.

candidates whose tickets went on to win show less residual, sentence-level pitch movement than candidates whose tickets did not. The pattern survives an unusually strong battery of measurement checks. It does not support reliable prediction for an unseen speaker or an unseen debate. And when the same prediction question is handed to classical statistical models and to a large language model, both appear to succeed for the same avoidable reason: speaker or transcript identity leaking into what should have been a held-out fold. This paper is organised around those three connected claims—a substantive discovery, a methodological diagnosis of why classical models mislead, and the same diagnosis run again on a language model—because that is the order in which the evidence actually has to be read: found, checked, tested for generalisation, and checked once more against the newest kind of instrument available for checking it.

A. What Debate Analysis Is

Debate analysis is the study of what happens when two people argue in public under fixed rules. It asks what was said, how it was said, and what effect it had. Traditionally this was done by hand: a researcher watched, transcribed, and coded. Computational debate analysis does the same work at scale, using programs that measure properties of the recording and the transcript, and it can therefore ask questions across decades of material rather than across a single evening.

The move to computation buys coverage and costs certainty. A human coder who marks a sentence as an appeal to patriotism can say why. A program that marks the same sentence has applied a rule, and whether that rule captures the idea is a separate question that usually goes unasked. Much of this paper is about what happens when it is asked.

Televised general-election debates are among the most watched political events in the United States. They reach tens of millions of viewers, they are one of the few occasions on which voters see the candidates side by side, and they are credited with shifting voter intention [1].

For a language researcher their appeal is different and more technical. Two candidates answer the same questions, in the same room, at the same time, before the same audience, recorded through the same equipment. Much of what normally makes spoken language hard to compare—topic, setting, interlocutor, recording chain—is held fixed by the format, which is

why we call this a *standardised naturalistic comparison* rather than a controlled experiment: nobody assigns which candidate stands on which side, and nobody assigns who wins. What varies, within a debate, is the two people.

The U.S. case adds three things. The debates have been held on a regular cycle since 1988, so a corpus can span decades with a consistent format. Transcripts and recordings are public. And each debate is followed, within weeks, by an election, which attaches an objective outcome to each speaker—something almost no other spoken corpus provides.

B. Why Language and Voice Must Be Examined Together

A spoken utterance can be described at several levels at once, and each answers a different question about the speaker.

Linguistic analysis asks how an argument is built. Counting clauses, measuring how deeply they nest, and measuring how far a word sits from the word it depends on are the standard ways to quantify syntactic complexity, the last with an established reading in terms of processing effort [2]. These are structural properties and barely depend on topic, which makes them comparable across debates about different things.

Semantic analysis asks what was said. Word lists give interpretable counts of sentiment and topic but detect vocabulary rather than meaning. Sentence embeddings capture meaning across paraphrase but are opaque. Word class and pronoun use indicate whether a speaker is talking about the people in the room or about things and ideas [3].

Acoustic analysis asks how it was said. Pitch and how much it moves describe intonation. Measures of the voice source describe whether a voice sounds clear, breathy or harsh [4]. Silent pauses indicate where planning effort falls [5], [6]. None of this survives transcription, so a text-only study discards it entirely.

A fourth layer, *discourse annotation*, labels what an utterance does—the act it performs and the rhetorical move it makes. This is where political discourse analysis has traditionally worked, and, as Section V shows, it is also the layer where the instrument matters more than the effect.

Three lines of prior work converge on the case for measuring all of this together rather than any one layer alone, and Section II covers them in detail: political communication research on what debates convey [1], speech science on the prosodic correlates of perceived charisma [7], [8], and computational text analysis on what word-level features carry [3], [9].

C. Gaps in Previous Research

Four gaps motivated this study.

Scale and span. Most multimodal debate studies examine one debate or one election cycle. Nobody has measured every general-election presidential debate over three decades on a common footing.

Modality separation. Acoustic studies rarely measure syntax; text studies discard the audio. Where a difference is found at one level, there is no way to know whether it is the same difference showing up twice.

Speaker–outcome confounding. A candidate’s speech and a candidate’s identity arrive together. Studies that compare groups of politicians rarely report how many distinct people underlie a reported difference, and a model trained to distinguish the groups may be learning to recognise the politicians rather than a general pattern of political communication.

Instrument validity. Keyword rules, acoustic descriptors and, more recently, language-model judgments are all used as measurements of theoretical constructs. The check that would discipline that use—does the measure covary with something external, and does it survive being asked why it answered that way?—is usually unavailable or simply unasked.

D. Research Questions

RQ1. Which linguistic, semantic, discourse and acoustic properties differ, within debates, between candidates whose tickets subsequently won and those whose tickets did not?

RQ2. What specific property of the pitch contour accounts for the strongest and most robust acoustic difference?

RQ3. Do the individual and combined measures that appear to distinguish the two groups generalise to debates involving previously unseen speakers?

RQ4. Can a large language model judge electoral outcome from debate material without relying on transcript memorisation or unsupported explanations?

One further question is exploratory rather than primary, because the corpus can only partly answer it: *for the small number of candidates observed in both a winning and a losing election cycle, do their speech characteristics change with the outcome?* Only George H. W. Bush and Donald Trump provide both conditions, and, as Section VII shows, their results disagree—so this question is reported as a case study (Section VII, Appendix H), not as evidence either way.

E. Contributions

- 1) A multimodal, sentence-aligned corpus of all 24 televised U.S. general-election presidential debates from 1988 to 2020, with an objective, externally determined outcome label attached to every candidate—whether their ticket won the election that followed. This label is not a measure of speaking well, and no claim in this paper depends on winning and speaking well being the same thing.
- 2) A common within-debate statistical screen of 43 linguistic, semantic, discourse and acoustic measures (RQ1), identifying which differences survive correction for having looked at 43 things.
- 3) A targeted characterisation of residual pitch control using ten higher-order and dynamic measures of the same pitch contour (RQ2), showing that the surviving difference is specifically about the movement left over once each sentence’s own trend is removed, not about speed, smoothness, or overall range.
- 4) A speaker-aware and contamination-aware evaluation protocol (RQ3, RQ4) showing how classical predictive models and a large language model can both appear to

generalise when they are in fact exploiting speaker or transcript identity—and the cheap checks that catch it in either case.

II. RELATED WORK

A. Computational Analysis of U.S. Political Debates

Benoit [1] establishes the substantive case: debates convey information about policy positions and personal character, and voters use it. This is a content-level result and says nothing about delivery, but it is why delivery is worth measuring. The broader literature on debate effects sits alongside longitudinal and rhetorical analyses of political speech more generally, and two properties of this literature shaped our design. First, studies are typically confined to one election or one debate, so nothing is known about whether a difference found in one cycle recurs in another. Second, and more importantly, the number of distinct speakers behind a reported result is rarely stated. In a corpus of political speech, speakers are few and memorable, and a difference between two groups of politicians may be a difference between two sets of individuals.

B. Prosody, Charisma, and Political Speech

The acoustic line begins with Rosenberg and Hirschberg [7], who identified prosodic correlates of perceived charisma in political speech: pitch dynamics, intensity range and pause structure. Their feature set is close to ours, and their finding motivated our expectation that pitch and pausing would matter most. Gravano et al. [8] extended the approach, showing that acoustic properties of political speech carry predictive signal for downstream behaviour. Two further strands of this literature are load-bearing here. Künzel [10] showed that average pitch is largely a fact about a speaker’s larynx rather than about their speech, the single most consequential constraint on how we measure pitch (Section V-C). Goldman-Eisler [5] and Levelt [6] tie pausing and, more generally, the moment-to-moment disruption of a planned utterance to the mechanics of speech production—the literature our residual-pitch mechanism (Section IX-B) draws on directly. Laver [4] supplies the taxonomy of voice quality that defines what harmonic-to-noise ratio and steadiness of delivery are meant to capture, on both the production and the perception side.

C. Multimodal Political-Speech Modelling

A separate strand combines transcript features, prosodic features, and sometimes learned embeddings into supervised models that predict an outcome of interest—perceived competence, persuasiveness, or, as here, an election result. This literature rarely asks the question that turns out to matter most for a corpus of famous, repeatedly-observed speakers: *does the model learn an outcome-associated communication pattern, or does it learn the identity of the recurring politicians who happen to carry each label?* A model can score well on held-out debates while still failing this test, if the speakers in those debates also appear elsewhere in training. Section VII makes this the central methodological question of the paper rather than a caveat appended to it.

D. LLMs as Annotators and Judges

Zheng et al. [11] popularised the language model as a judge of open-ended output. Wei et al. [12] showed that prompting a model to reason step by step before answering can improve performance on tasks with a checkable right answer; whether the same holds for a subjective coding scheme, where no answer key is fully reliable, is an open question we test rather than assume (Section V). Magar and Schwartz [13] showed that a model’s apparent competence on a task can be memorisation of the evaluation material rather than reasoning over it—a concern that is not hypothetical for this corpus. Presidential debate transcripts are among the most reproduced texts in any model’s training data, so Section VIII treats this not as a caveat to mention but as something to measure directly, with a contamination probe and a closed-vocabulary grounding check rather than trust in a model’s stated reasons.

E. Positioning of the Present Study

Prior research suggests that linguistic and prosodic properties carry political and interpersonal information, but it remains unclear which differences persist across election cycles, whether they generalise beyond familiar speakers, and whether LLM judgments reflect evidence-based reasoning rather than recognition of historically prominent transcripts. This paper answers all three for one corpus, under one shared statistical and measurement-validity protocol (Section IV).

III. CORPUS CONSTRUCTION AND DATA QUALITY

A. Corpus Scope and Inclusion Criteria

The corpus covers every televised general-election presidential debate held in the United States between 1988 and 2020: 24 debates across nine election cycles. This is a complete enumeration for the period rather than a sample. Vice-presidential debates are excluded because consistent transcripts do not exist for all cycles.

For each debate we collected the broadcast audio and the official transcript, and paired them by election year and debate number. Election-outcome metadata supplies one record per candidate per debate, carrying party affiliation and whether that candidate’s ticket won the election that followed. Appendix A lists every debate, its date, its candidates, and the outcome attached to each.

Two further debates, from 2024, were collected but are held out of every analysis reported here, because their speaker attribution could not be verified: their official transcripts are paraphrased news summaries sharing only 5.2% and 5.5% of their words with the audio. Appendix I documents the evidence and reports what changes if they are included.

B. Transcript Preprocessing

Matching text to audio and analysing language need opposite things. Matching needs a regular token stream, so we build a stripped version: lower case, no punctuation, contractions left alone. Analysis needs the opposite, because sentence boundaries, part-of-speech tags and dialogue acts all depend on capitals and punctuation. We keep both and never analyse the stripped one. Stage directions and applause markers resolve to no speaker and are dropped before sentence segmentation.

TABLE I
EMPIRICAL STUDIES THIS WORK BUILDS ON MOST DIRECTLY. “GAP ADDRESSED HERE” IS WHAT SECTIONS IV–VIII DO THAT THE CITED STUDY’S DESIGN COULD NOT.

Study	Modality	Unit	Task	Finding	Gap addressed here
Benoit [1]	Content/text	Debate	Effects on voters	Debates shift information voters use	No delivery layer; single-cycle scope
Rosenberg & Hirschberg [7]	Acoustic	Utterance	Perceived-charisma correlates	Pitch, intensity, pausing correlate with charisma	No election outcome; no cross-speaker holdout
Gravano et al. [8]	Acoustic	Speech sample	Predict downstream behaviour	Acoustic features carry predictive signal	No multi-decade span; speaker count unreported
Tausczik & Pennebaker [9]	Text	Word	Word-counting method	Closed-class words carry social/psych. signal	Text only; no acoustic pairing
Pennebaker [3]	Text	Word	Function-word signal	Function words outweigh content words	Same as above
Künzel [10]	Acoustic	Speaker	Pitch–anatomy relationship	Mean F0 mostly reflects the larynx, not the speech	Motivates variation-not-level measurement here
Goldman-Eisler [5]	Acoustic	Utterance	Pausing and planning	Pausing indexes planning, not just rate	No election-outcome application
Levelt [6]	Theoretical	Utterance	Production model	Planned vs. online formulation differ in execution	Motivates residual-pitch mechanism (Sec. IX-B)
Gibson [2]	Text	Sentence	Dependency-length cost	Longer dependencies cost more to process	One of 43 syntactic measures here
Biber [14]	Text	Corpus	Register/dimension recovery	Registers need many co-occurring features	Standard our embedding axis fails (Sec. V-A)

C. Speaker-Label Normalisation

Speaker labels vary across three decades and even within one transcript (MR. BUSH, BUSH:, THE PRESIDENT). Each is matched against the roster for that debate and resolved to one of four roles: candidate (with party and outcome), moderator, audience, or unattributed.

D. Audio–Text Alignment

Broadcasts include studio commentary before and after, which is not part of the debate and would corrupt both the audio measurements and the speaker list. We locate the start and end by finding the opening and closing formulae in the audio, and time everything from there. Figure 1 summarises the full pipeline from raw audio and transcript to the aligned corpus used in every analysis below.

Every acoustic measurement needs to know which piece of audio corresponds to which sentence. This is the hardest practical step, because the transcripts are not equally faithful: some were produced verbatim, others tidied for readability. Rather than assume, we measure. For each debate we run speech recognition on the audio and count how many words of the official transcript also appear, in order, in the recognised text. This match rate is computed before any decision is taken and determines how the debate is handled (Table II). Where the match is high the official transcript is authoritative: shared words take their timings directly and the gaps are filled by interpolation. Where it is lower, the same method applies with more interpolated timing.

Forced alignment was also run and worked on 20 of the 24 debates. We still used the recognition-based method throughout, for a design reason. If 20 debates took timings from one method and four from another, the corpus would carry a difference in timing accuracy that lines up with the age of the recording, and therefore with the period—exactly the confound the analysis is built to remove. One method

TABLE II
HOW THE TRANSCRIPT WAS MATCHED TO THE AUDIO, DECIDED BY THE MEASURED MATCH RATE BEFORE ANY ANALYSIS.

Treatment	Debates	Sentences	Match rate
High fidelity	22	25,693	88.4–93.3%
Reduced fidelity	2	3,979	69.7–82.3%
Total	24	29,672	—

everywhere avoids it. The forced alignment output is kept as an independent check (Appendix B).

Each sentence then receives measurements at three levels—linguistic, semantic, and acoustic, plus a discourse-tag layer—detailed in Section IV-B. Tools, versions and parameter settings are in Appendix C.

E. Quality Control and Exclusions

Of the 24,193 candidate sentences, 23,109 (95.5%; 10,868 winning, 12,241 non-winning) carry complete acoustic measurements. The 1,084 missing are 846 sentences shorter than the 300 ms extraction floor plus 238 extraction errors. Missing data is even across the two outcome groups (4.3% against 4.7%) but very uneven across alignment treatments (2.8% against 16.5%)—one of several reasons every statistical model below fits one term per debate (Section IV-D), which removes any difference between debates, including this one, from the comparison of interest. No two candidate sentences overlap in time, so interruptions and overlapping speech are not represented anywhere in the corpus. Appendix I lists the remaining known data-quality issues (near-duplicate sentences, zero-duration rows, extraction failures) in full.

F. Final Corpus Statistics and Speaker Structure

The corpus holds 29,672 time-aligned sentences (Table III), of which candidate sentences—the analytical population—are

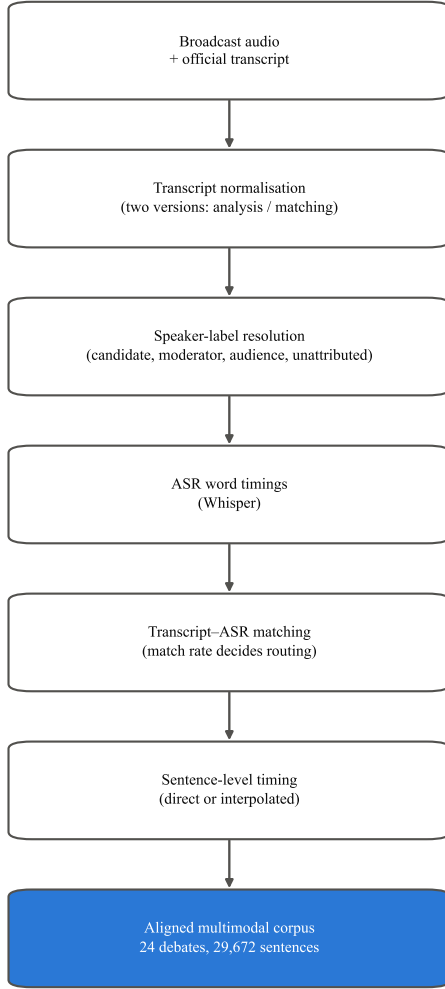


Fig. 1. Corpus construction pipeline. The match rate between the official transcript and recognised speech, computed before any analytical decision, determines whether a debate is routed to direct timing or interpolation (Table II).

TABLE III
WHO SPEAKS IN THE CORPUS.

Speaker role	Sentences	Share
Candidate, winning	11,355	38.3%
Candidate, non-winning	12,838	43.3%
Moderator	5,192	17.5%
Audience	287	1.0%
Total	29,672	100%

81.5%. Table IV gives the breakdown by election. The two groups differ in size because candidates get different amounts of floor time and because in 1992 three people shared the stage, two of whom lost. This does not affect the analysis, since every debate contributes both groups and all comparisons are made within debates.

One statistic is not in either table and governs the interpretation of every result in this paper: these 24,193 sentences come from 14 **people**. Twelve of them contested a single election, so they appear in one outcome group only. This is not a minor

TABLE IV
CANDIDATE SENTENCES BY ELECTION CYCLE.

Cycle	Debates	Winning	Non-winning
1988	2	821	770
1992	3	832	2,058
1996	2	816	1,270
2000	3	1,625	1,192
2004	3	1,458	1,569
2008	3	1,189	1,302
2012	3	1,227	1,733
2016	3	2,035	1,250
2020	2	1,352	1,694
Total	24	11,355	12,838

TABLE V
WHO IS IN THE CORPUS. TWELVE OF FOURTEEN CANDIDATES APPEAR IN ONLY ONE OUTCOME GROUP; TWO APPEAR IN BOTH.

Speaker	Debates		Sentences	
	Win	Lose	Win	Lose
Donald Trump	3	2	2,035	1,694
George W. Bush	6	0	3,083	—
Barack Obama	6	0	2,416	—
George H. W. Bush	2	3	821	1,034
Bill Clinton	5	0	1,648	—
Mitt Romney	0	3	—	1,733
John Kerry	0	3	—	1,569
Joe Biden	2	0	1,352	—
John McCain	0	3	—	1,302
Bob Dole	0	2	—	1,270
Hillary Clinton	0	3	—	1,250
Al Gore	0	3	—	1,192
Ross Perot	0	3	—	1,024
Michael Dukakis	0	2	—	770
Total	24	24	11,355	12,838

limitation stated in passing; Sections VII and IX-C show in detail why it governs whether any predictive result can be trusted. Table V gives the structure.

IV. ANALYTICAL AND EXPERIMENTAL FRAMEWORK

A. Outcome and Unit of Analysis

Every candidate–debate record carries a binary outcome label, defined as follows:

$$w_{d,s} = \begin{cases} 1, & \text{ticket } s \text{ won the election after debate } d, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

We use three units of analysis. *Sentence level* is the finest grain, used for the statistical screen (Section IV-D) and for training classifiers (Section VII-B). Measurements exist at this level, but the outcome label does not: it is assigned once per candidate–debate, so all sentences from one candidate in one debate share one label. *Candidate-by-debate* is the aggregate used wherever speaker identity itself is the variable of interest (Table V; Section VII). *Debate-level paired prediction* asks, for a whole debate, whether the model’s aggregate score ranks the winning candidate above the opponent. Every generalisation claim in Sections VII and VIII is judged at this level, because it is the only one of the three not affected by class imbalance or by dependence between sentences from the same speaker.

B. Broad Multimodal Measure Set

We measure each sentence at four levels, 43 measures in total; the complete dictionary is in Appendix C. *Linguistic* measures are clause count, subordinate-clause count, parse depth, mean dependency length, sentence type, word length, vocabulary richness, the balance of function to content words, part-of-speech proportions, four readability scores, and pronoun rate by grammatical person. *Semantic* measures are sentiment (overall, positive, negative parts), membership in 35 topic categories fixed in advance, and a sentence embedding. *Discourse* measures are dialogue acts and rhetorical categories, under a scheme extending DAMSL [15] with categories for political rhetoric, labelled independently by rule and by a large language model (Appendix E). *Acoustic* measures are average pitch, its variation, range and slope, computed both in hertz and on a semitone scale (a hertz-based measure of variation grows with average pitch automatically, so the semitone scale is needed for a fair comparison across speakers); loudness and its variation; three measures of the voice source, including harmonic-to-noise ratio computed by two independent tools; silent-pause count, total duration and average duration; and speech and articulation rate in words and syllables.

C. Targeted Pitch-Contour Measure Set

Section V-C shows that narrower semitone pitch range is the one acoustic property that survives every validity check in the broad screen. Every measure up to that point treats pitch variability as a single number per sentence—a standard deviation, or the width of a percentile band. Such a number answers *how much* a speaker’s pitch moves and says nothing about the *shape* of that movement. We therefore extract the pitch contour itself and build ten further measures of its shape and dynamics, as a targeted follow-up on the one construct already established by the broad screen.

For each candidate sentence we recover the frame-level semitone-scaled pitch contour—the same quantity summarised by the surviving broad-screen measure, but before it is collapsed to one number—using the openSMILE eGeMAPSv02 low-level descriptor F0semitoneFrom27.5Hz_sma3nz at its native 10 ms frame rate, extracted independently per sentence clip to match the original per-clip procedure exactly. We validate this extraction against the already published semitone functionals for 1,000+ sentences and find the two agree to within floating-point rounding (mean absolute difference 0.0006). A sentence needs at least 10 voiced frames (100 ms of voiced speech) to be included; this is a floor rather than a guarantee of reliability, so we additionally assess reliability by stratifying on duration after the fact, the same way we do for voice clarity in Section V-C.

From the voiced frames of each sentence we compute ten measures in two groups. *Distributional shape* (order-independent): *skewness*, whether the distribution leans toward higher or lower pitch with a longer tail on one side; *kurtosis*, whether frames cluster near a central value with occasional large excursions, or spread more evenly across the range; and the *interquartile-to-range ratio*, the width of the middle half of the distribution relative to its 20–80 percentile span.

Dynamics (order-dependent; computed only across pairs of consecutive voiced frames, so no estimate spans a silent or unvoiced gap): *pitch velocity*, the mean and standard deviation of the frame-to-frame semitone change; *pitch acceleration*, the mean absolute second difference; *turning points per second*, how often the contour changes direction; *total variation*, the summed absolute frame-to-frame movement normalised by voiced duration; and, after fitting a linear trend to each sentence’s contour and removing it, the *declination slope* of that trend and the *declination-residual standard deviation*—the spread of whatever pitch movement is left once the sentence’s overall rise or fall is subtracted out. We evaluate all ten with the statistical design of Section IV-D, applying Benjamini–Hochberg correction at $q = 0.05$ across this family of ten measures, kept separate from the broad-screen family of 43.

D. Within-Debate Statistical Comparison

Every analysis in this paper, broad screen and targeted follow-up alike, uses one statistical design, described once here.

Comparing within debates. For each measure y and sentence i , with $w_i = 1$ for winning candidates, we fit

$$y_i = \beta w_i + \sum_d \alpha_d D_{i,d} + \varepsilon_i, \quad (2)$$

where $D_{i,d}$ marks sentence i as belonging to debate d . One term per debate removes everything that varies between debates, so β is the difference *inside* debates: free of period, format, equipment, moderator and topic. We report $\beta/\sigma_{\text{within}}$, the difference in standard deviations.

Uncertainty and p-values. We compute cluster-robust standard errors, treating each debate as a cluster and, separately, treating each speaker as one. Every effect carries a 95% confidence interval. Our primary p -value is an exact randomization test: we flip the winner label within each debate and recompute the effect; with 24 debates there are $2^{24} = 16,777,216$ possible flips, and we enumerate all of them.

Correcting for multiple measures. With 43 measures, about two would clear $p < 0.05$ by chance alone. We control the false discovery rate at $q = 0.05$ with the Benjamini–Hochberg procedure [16]: we sort the p -values smallest first and compare the k -th against $k \times 0.05/43$.

Asking whether a result would repeat. We pool the 24 debates with a random-effects model and report how much the effect varies between debates (I^2), a 95% confidence interval for the average, and a 95% *prediction interval* for a new debate. The confidence interval answers what the average difference was across the debates we have. The prediction interval answers whether we would see the difference at all in a debate we have not.

E. Measurement-Validity Protocol

Significance and multiple-comparison correction are not sufficient grounds to call a measure reliable. We therefore require every candidate measure in this paper to clear seven further checks before it is reported as reliable rather than merely detected:

- 1) **Speaker coupling.** Does the measure correlate with a speaker’s own habitual pitch, across speakers and within a single speaker’s own debates (Sections V-C, VI)?
- 2) **Cross-implementation agreement.** Does it agree across independent tools measuring the same quantity (Praat versus openSMILE; Section V-C)?
- 3) **Duration reliability.** Is it reliable across the range of sentence durations actually present in the corpus (Table XIV)?
- 4) **Alignment-treatment stability.** Does it hold sign under both the high- and reduced-fidelity alignment treatments (Table II)?
- 5) **Leave-one-speaker-out stability.** Does it retain direction when each speaker in turn is removed and the effect recomputed?
- 6) **Cross-debate recurrence.** Does it appear in a majority of the 24 debates individually, not only in the pooled average?
- 7) **Prediction-interval support.** Does its 95% prediction interval for a new debate exclude zero?

Section V-C reports the largest observed effect in the paper failing checks 1–2. Section VI reports the primary finding passing checks 1–6 and failing check 7; we treat this as the paper’s central qualification rather than a caveat appended to it.

F. Generalisation Protocol

We use two evaluations throughout, and they answer different questions. **Debate held out:** the debate under test is absent from training, but its speakers’ other debates may remain. This is the standard setup in most published work. **Speakers held out:** every debate containing either of the test debate’s speakers is absent from training, so neither candidate’s voice appears anywhere in the model. We treat the second as the primary evaluation of generalisation in this paper and report the first alongside it, because the gap between the two is itself the result we report in Section VII.

G. LLM Evaluation Protocol

For each of the 24 debates we build an anonymised record: every candidate sentence with names, party and the date removed, each candidate relabelled *Candidate A, B, or C* in an order re-randomised on every repeat. We ask the model (Anthropic’s `claude-opus-4-8`; full specification in Appendix G) which candidate went on to win that year’s election, under two conditions built from the same 34 of the 43 measures used in Section VII-A (those with complete values on enough rows): **transcript** (anonymised candidate turns, in order, and nothing else) and **descriptors** (the 34 within-debate standardised measures, averaged per candidate, as a table, with no transcript text at all). We also run a third condition, both together; it agrees with the transcript-only condition on all 24 debates, so we report it only in Appendix F.

Every call also requires a short explanation and up to five cited reasons, each drawn from a closed vocabulary—the 34 measure names for the descriptor condition, eight qualitative

themes for the transcript condition—so that a citation can be checked by machine against the real data rather than read and judged by hand. We run two further checks alongside the prediction itself. A **contamination probe** sends the model the same anonymised input, with no prediction question at all, and asks it to identify the specific debate. A **grounding check** compares every cited measure against the real standardised value for that debate. We score each debate three times per condition, re-randomising the candidate letters each time.

V. BROAD MULTIMODAL SCREENING

This section answers RQ1: which properties differ, within debates, between winning and non-winning candidates, across 43 measures at four levels.

A. Linguistic and Semantic Results

Before reading any result from the subordination counter, one property of it must be stated: it accepts dependency relations that admit non-finite complements, and reports 1.41 subordinate clauses out of 1.84 total in a fourteen-word sentence. This is too high to be real subordination, so we read this measure as “clausal material” rather than subordination.

Sentence structure. The pattern here is weak and points one way. Winning candidates produce slightly more clausal material: subordinate clauses +0.093, clause count +0.086, parse depth +0.029, sentence length +0.034. None of these effects survives correction, and every confidence interval includes zero. The effects are also two to eight times larger in the reduced-fidelity treatment, which places most of them in the two debates from one cycle rather than across the corpus.

Words and semantics. Adjective rate is the strongest single text measure, at +0.062 ($p = 0.0036$); it misses correction by a hair, and clears in any family of 41 measures or fewer but not in 43. We report it without explanation, since we had no prior expectation about adjectives. Pronouns do not separate the groups at all: second person -0.048 , third person +0.009, first person -0.003 ; nor do readability or sentiment (+0.030, $p = 0.27$). We record the pronoun null on its own, because in a debate *you* nearly always means the opponent: if these two groups differ, it is not in how much they address each other.

The embedding axis. We draw the line between the two groups’ average sentence vectors and sort the corpus along it. This produces a striking result: the 50 highest-scoring sentences split 40 to 10 between winning and non-winning candidates, a stronger separation than anything else in this section. An examination of these sentences shows that it is not a separation between the groups. Among the 200 highest-scoring sentences, 47.5% contain *she*, *her* or *hers*, against 1.9% in the corpus overall; 32 of the top 50 come from one candidate and 29 from the 2016 cycle. The axis separates talking about a female opponent from talking about past presidents, in one election. It is not a sentence-length axis either (correlation +0.09; removing length leaves the poles unchanged). We report this as a diagnostic of what a single difference-of-centroids direction can produce by accident, not as a finding: following Biber [14], a register dimension would have to be recovered from many co-occurring features across texts.

B. Discourse-Measure Validity

This layer produced no usable result, and the reason lies in the instrument rather than the underlying construct. We check what fraction of each category’s hits come from a single string, and find that several of the rule-based detectors are keyword matchers rather than concept detectors: unity framing and patriotic appeal are dominated by the strings *United States* and *America/American*; historical reference by a bare four-digit year; fear appeal by unbounded substring matches. Fixing the four fixable rules removes 94–97% of what these categories had counted, and what survives appears in under 0.5% of sentences with differences no larger than 0.014. A second, independent labelling by a large language model agrees with the rule-based one at chance on the interpretive, rhetorical categories (Krippendorff’s $\alpha = +0.012$) and only modestly better on the more grammatically observable dialogue acts ($\alpha = +0.199$); a chain-of-thought variant of the same prompt does not close this gap. We draw no conclusion about rhetorical strategy from this layer anywhere in this paper. The full trigger audit, category-level agreement table, and chain-of-thought comparison are in Appendix E.

C. Acoustic Results

Following Künzel [10], we treat average pitch as weak evidence when comparing different people, and treat variation as more meaningful, since variation describes what a speaker does rather than what their larynx is. Any measure that correlates with a speaker’s own average pitch is therefore suspect, and we test each one for it.

The two measures that survive. The semitone pitch range between the 20th and 80th percentiles averages 4.51 for winning candidates and 4.91 for non-winning ones: winners move their pitch across a narrower band. The difference is -0.226 standard deviations, confidence interval $(-0.332, -0.120)$, randomization $p = 0.00017$, present in 21 of 24 debates. Dropping any one speaker leaves it between -0.308 and -0.169 , always the same sign. It holds in both alignment treatments (-0.229 and -0.206), barely correlates with a speaker’s own average pitch ($r = +0.11$), and correcting for average pitch makes it *stronger* (-0.267). The second measure, pitch variation relative to average pitch, is larger (-0.332 , $p = 8.8 \times 10^{-6}$, 20/24) but less clean: it correlates with a speaker’s average pitch at $r = -0.49$ and collapses to $+0.021$ in the reduced-fidelity treatment. We treat the percentile range as primary and the ratio as corroborating, a choice made on measurement quality rather than on p -values. Both measures survive Benjamini–Hochberg correction and would survive the stricter Bonferroni threshold.

Voice clarity: the largest effect, and what it measures. The biggest single effect in the study is harmonic-to-noise ratio: $+0.461$ on one tool, $+0.315$ on the other. At face value, winners have clearer voices. The validity checks say we cannot tell what it measures. Across the fourteen speakers, average pitch correlates with average harmonic-to-noise ratio at $r = +0.776$ on one tool and $+0.896$ on the other: ranking the speakers by voice clarity nearly reproduces the ranking by pitch, most likely because a fixed 60 ms analysis window

TABLE VI
THE TEN PITCH-CONTOUR MEASURES. EFFECT IS $\beta/\sigma_{\text{WITHIN}}$; DIR. IS DEBATES IN THE POOLED DIRECTION OF 24; I^2 IS IN PER CENT; PI IS THE 95% PREDICTION INTERVAL. ORDERED BY EXACT RANDOMIZATION p .

Measure	Eff.	p_{perm}	Dir.	I^2	LOSO	stable	BH
Declination-residual SD	-0.242	0.0005	20/24	96		yes	•
Skewness	-0.265	0.0014	19/24	97		yes	•
IQR / 20–80 ratio	-0.053	0.0036	16/24	24		yes	•
Turning points / sec	$+0.039$	0.431	12/24	93		no	–
Declination slope	$+0.014$	0.492	14/24	65		no	–
Pitch acceleration	-0.023	0.667	14/24	92		no	–
Pitch velocity (mean)	-0.025	0.668	13/24	93		no	–
Total variation	-0.025	0.668	13/24	93		no	–
Kurtosis	-0.019	0.796	14/24	97		no	–
Pitch velocity (SD)	$+0.013$	0.881	14/24	97		no	–

gives a low-pitched voice fewer cycles to work with, and because television transmission cuts low frequencies. The two tools then behave differently under a within-speaker check: in 51 speaker-by-debate cells the correlation collapses to a median of $+0.083$ for one tool but stays at $+0.566$, positive in all 51 cells, for the other. Two standard tools for the same quantity should not disagree this way if both are measuring voice quality. We take this as the clearest single illustration in the paper of Section IV-E’s checks 1–2: same family, same corpus, same statistics as the surviving pitch-range measure, and this one fails both.

Pausing. Winners pause more on every version of the measure ($+0.14$ to $+0.20$ across five variants), consistent across alignment treatments and duration bins. The speaker check rules this out: **all six pause measures change sign when Barack Obama is removed**, from $+0.196$ to -0.021 for pause proportion; his three 2012 debates alone carry effects of $+0.96$ to $+1.25$. The measure itself is sound; the finding is a description of one person, not of winning candidates in general.

Figure 2 shows all 43 measures together.

VI. RESIDUAL PITCH CONTROL: THE PRIMARY FINDING

This section answers RQ2.

A. From Narrower Pitch Range to Residual Movement

We build the result in four steps. First, the broad screen (Section V-C) shows winning candidates use a narrower semitone pitch band. Second, that number alone does not say what kind of movement produces the narrower band: a flatter overall melody and a steadier local wobble on top of the same melody would both shrink it. Third, the frame-level measure set of Section IV-C asks the question directly, by removing each sentence’s own linear rise or fall and measuring what is left. Fourth, what is left is smaller for winning candidates: a specifically *local*, moment-to-moment difference in pitch control, not a difference in overall prosodic sweep. Figure 3 illustrates the distinction with two synthetic contours that share one sentence-level sweep and differ only in residual movement.

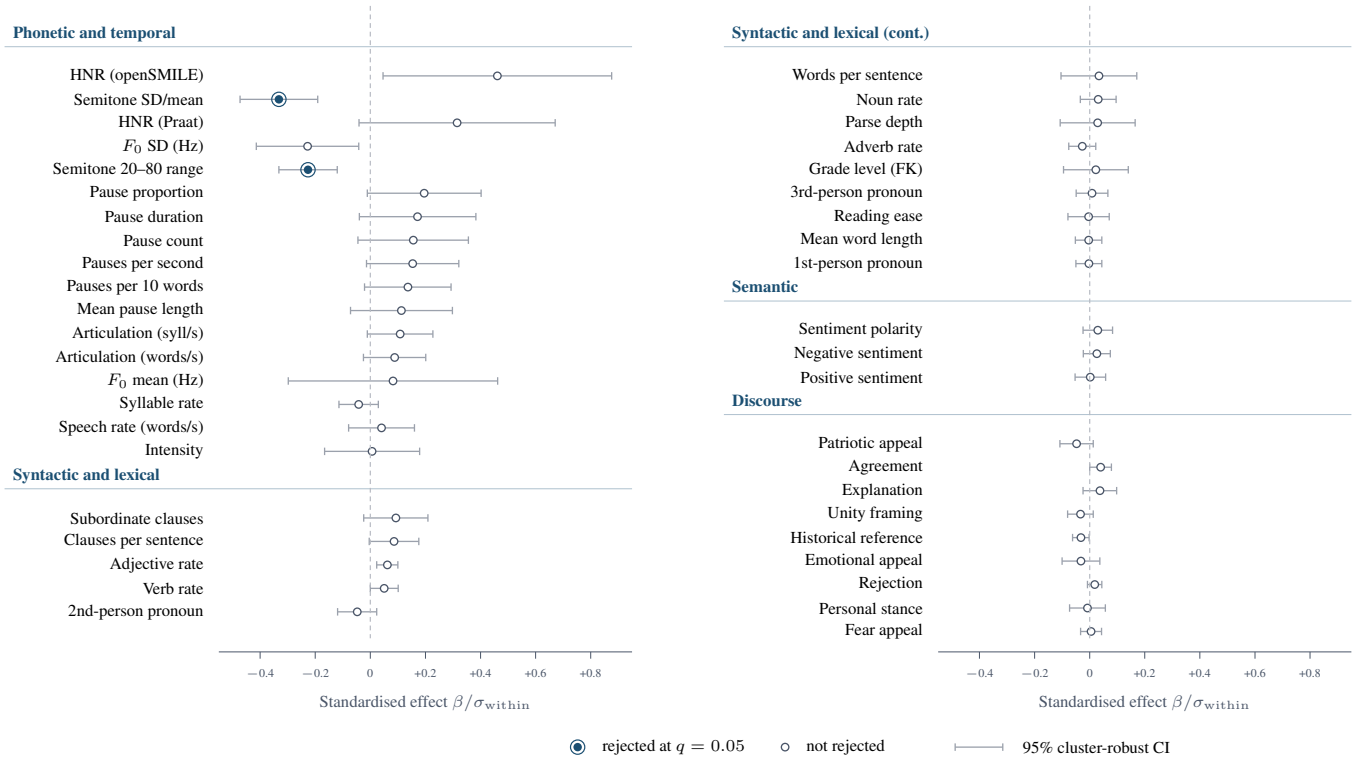


Fig. 2. Within-debate difference for all 43 measures, in standard deviations, with 95% confidence intervals that account for clustering by debate. Positive means higher among winning candidates. Filled markers are the two measures that survive correction for multiple comparisons. Note the scale: the biggest effect is 0.46 and most are under 0.05.

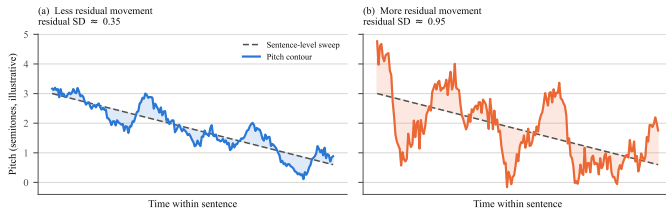


Fig. 3. What “residual pitch movement” means. Both panels share the same sentence-level sweep (dashed); the shaded area is the residual left over once that sweep is subtracted out. Synthetic illustration, not real corpus data.

B. Primary Result

Table VI and Figure 4 show that three measures clear correction and seven do not. **Declination-residual standard deviation is the cleanest result in the paper.** Winning candidates have less pitch movement left over once each sentence’s own linear rise or fall is removed: effect -0.242 , exact randomization $p = 0.0005$, present in 20 of 24 debates (sign-test $p = 0.0015$), sign-stable across all fourteen leave-one-speaker-out refits. It passes the two checks that separated the broad screen’s two surviving measures from each other (Section V-C): correlation with a speaker’s own mean pitch is $r = -0.06$ across speakers and $r = +0.23$ within speaker-by-debate cells, uncoupled by either reckoning, unlike the confounded voice-clarity measure. Rather than shrinking with sentence length, the way a noise artefact would, it *grows*: from -0.17 in sentences under a second to -0.53 in sentences over six. It holds sign in both alignment treatments, more weakly in

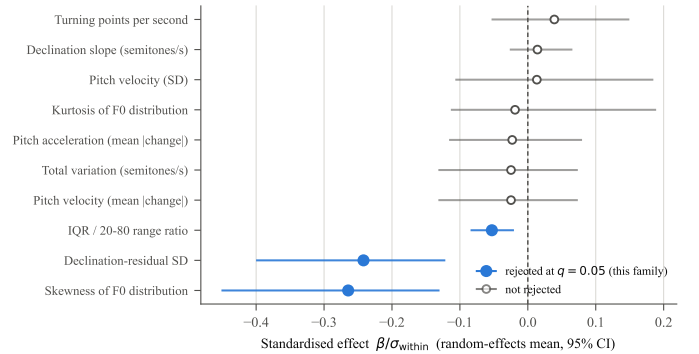


Fig. 4. Within-debate standardised effect (random-effects mean, 95% confidence interval) for the ten higher-order and dynamic pitch measures. Filled markers survive Benjamini–Hochberg correction within this family.

the reduced-fidelity group (-0.26 against -0.08) but without reversing.

C. Secondary Pitch-Shape Findings

Skewness survives the same correction and fails the same check that demoted the broad screen’s secondary pitch measure: it correlates with a speaker’s own mean pitch at $r = -0.29$ across speakers and $r = -0.24$ within cells, and reverses sign under alignment treatment (-0.31 against $+0.08$). By the same standard applied throughout, we report it as corroborating rather than clean. *The interquartile-to-range ratio* has the lowest heterogeneity of any measure in this

paper ($I^2 = 24\%$ against 94–97% elsewhere), and is stable across duration bins and alignment treatment. Its per-debate sign count, however, is the weakest of the three (16/24, sign-test $p = 0.15$), and it shows the same across-speaker pitch coupling as skewness ($r = -0.26$). We report it as secondary: consistent in magnitude whenever it appears, but appearing less consistently than either of the other two.

D. What the Null Results Rule Out

Pitch velocity, acceleration, total variation and turning-point rate—every measure of how quickly or how erratically the contour moves—show no difference between the groups, with I^2 above 90% throughout and prediction intervals spanning zero by a wide margin. Kurtosis and the declination slope itself show none either. Taken with the positive result above, this rules out an entire class of explanation: it is not that winning candidates move their pitch faster, more smoothly, or less erratically. It is specifically that the *spread* of where their pitch sits, and, more precisely, the spread that remains after removing each sentence’s own trend, is smaller. The data therefore support a distributional account of the effect rather than a kinematic one.

VII. GENERALIZATION AND SPEAKER CONFOUNDING

This section answers RQ3. We treat it as a central result rather than a limitation appended to one.

A. Interpretable and Nonlinear Models

If text and audio carry different information, combining them should carry more than any single measure. We use two models that make different assumptions about how measures combine: a linear discriminant, which uses the full covariance structure, and a decision forest, which imposes no shape on the data at all. Both use the same 34 of the 43 measures with complete values on enough rows, over 23,082 sentences, standardised within debate first so only within-debate contrasts contribute:

$$z_{ij} = \frac{x_{ij} - \mu_{dj}}{\sigma_{dj}}, \quad (3)$$

We also fit a simple weighted sum, each measure weighted by how well it separates the groups in training debates and summed with no learned covariance, and retain it only as a baseline row in Table IX, since it never outperforms either structured model. Full method and results for principal component analysis, which asks whether the outcome happens to line up with the largest-variance directions in the same 34 measures without ever seeing the labels, are in Appendix F. In short, it does not: the largest components are a sentence-length axis and a speaking-rate axis, neither of which separates the groups, and speaker identity explains 83–87% of a candidate’s position on the two largest components.

B. Feature Importance

We fit a random forest of 400 trees, maximum depth 8, minimum 50 sentences per leaf, drawing $\sqrt{34}$ measures at random for each split, class weights balanced. Importance is

TABLE VII
TOP TEN MEASURES BY PERMUTATION IMPORTANCE IN THE DECISION FOREST, UNDER BOTH HOLDOUT SCHEMES. NINE OF THE TEN ARE ACOUSTIC UNDER EITHER SCHEME.

Measure	Debate held out	Speakers held out
1 Voice clarity (openSMILE)	0.0308	0.0546
2 Semitone pitch SD/mean	0.0230	0.0526
3 Pitch SD (Hz)	0.0069	0.0092
4 Pause duration	0.0041	0.0033
5 Loudness	0.0040	0.0057
6 Semitone 20–80 pitch range	0.0036	0.0136
7 Pitch range (Hz)	0.0033	0.0056
8 Voice clarity (Praat)	0.0017	0.0106
9 Mean pitch	0.0014	—
10 1st-person pronouns	0.0009	—

Best text measure: 1st-person pronouns, 0.0009—about 3% of what the top acoustic measure contributes.

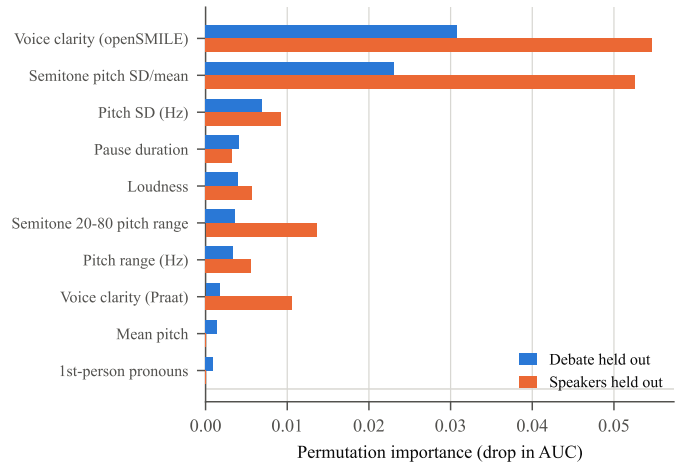


Fig. 5. Permutation importance of the top ten measures. The ranking is almost entirely acoustic, and it is led by the two measures shown in Section V-C to be tied to the speaker’s own voice.

permutation-based: after fitting, each measure is shuffled in the held-out data and we record how far the area under the curve falls—more trustworthy than split counts, which favour measures with many distinct values. We fit it twice, holding out one debate at a time and, separately, holding out every debate containing either held-out speaker.

Table VII and Figure 5 show that nine of the top ten measures are acoustic under either holdout scheme; the best text measure contributes about three per cent of what the top acoustic measure does. The forest leans hardest on the two measures already shown to be compromised—voice clarity and semitone pitch variation, both tied to the speaker’s own voice (Section V-C)—while the measure that survives every validity check ranks only sixth. The ranking barely changes when speakers are held out. What changes, in the next section, is whether the ranking is worth anything.

C. Debate Held Out Versus Speakers Held Out

We use the same random forest and the same two holdout schemes to answer the practical question directly: given a

TABLE VIII
HOW OFTEN THE FOREST IDENTIFIES THE WINNING CANDIDATE. ALWAYS GUESSING THE LARGER GROUP SCORES 0.530.

	Debate held out	Speakers held out
<i>Sentence level</i> (23,082 sentences)		
Accuracy	0.596	0.518
Area under curve	0.630	0.523
<i>Debate level</i> (24 debates)		
Debates called correctly	17/24	12/24
p against a coin	0.064	1.00
Majority-class baseline	0.530	

TABLE IX
THE PAIRED TEST. FOR EACH DEBATE, DID THE WINNING CANDIDATE'S MEAN SCORE BEAT THE OPPONENT'S?

Model	Held out	Debates	p
Weighted sum	debate only	17/24	0.064
	both speakers	11/24	0.84
Discriminant	debate only	19/24	0.0066
	both speakers	10/24	0.54
Decision forest	debate only	17/24	0.064
	both speakers	12/24	1.00

sentence, or a whole debate, how often can a model say which candidate went on to win?

Table VIII shows that with the debate held out, sentence accuracy is 0.596 against a 0.530 baseline and debate-level calls are 17/24, not significant against a coin ($p = 0.064$). With the speakers held out, sentence accuracy falls to 0.518, *below* the majority-class baseline, and debate-level calls are 12/24, an exact coin flip. Errors stay symmetric under speaker holdout (precision/recall near 0.49/0.54 either way), which indicates an uninformative model rather than a biased one.

The natural test of speaker generalisation—hold out a speaker, predict their sentences, score the result—cannot be run directly here: twelve of fourteen speakers appear in only one outcome class, so within those folds there is only one class to discriminate and the area under the curve is undefined, and pooling scores across folds compares differently-calibrated models rather than measuring discrimination within any one of them. The valid test is paired and within debate: for each debate, does the winning candidate's mean score beat the opponent's? Both classes are present by construction and both are scored by the same model.

Table IX and Figure 6 give the cleanest result in the paper. Holding out only the debate, the discriminant calls 19 of 24 correctly ($p = 0.0066$), and the weighted sum and forest call 17 of 24 each ($p = 0.064$). Holding out the speakers as well, the discriminant falls to 10/24, the forest to 12/24, and the weighted sum to 11/24. Three methods, three different assumptions about the shape of the data, one result: what looked like a model of how winners speak was a model of who the winners were.

D. Predictive Association Versus Recurrence

Every prediction interval in Table X includes zero, for every measure that survived every check up to this point. A

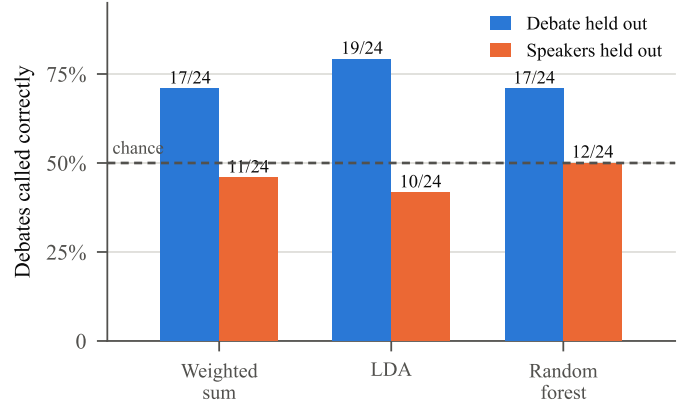


Fig. 6. The same three methods, evaluated two ways. Holding out the debate leaves both of its speakers in the training data; holding out the speakers removes them entirely. All three fall to chance under the stricter test.

TABLE X
REPEATABILITY. "DIR." IS DEBATES IN THE SAME DIRECTION OUT OF 24; I^2 IS THE PERCENTAGE OF VARIATION BETWEEN DEBATES RATHER THAN SAMPLING NOISE; LOSO IS THE RANGE WHEN EACH SPEAKER IN TURN IS DROPPED; PI IS THE PREDICTION INTERVAL FOR A NEW DEBATE.

Measure	Eff.	Dir.	I^2	LOSO	PI
Semitone 20–80 range	−0.226	21/24	94	[−.31, −.17]	(−.81, +.32)
Semitone SD/mean	−0.332	20/24	96	[−.44, −.27]	(−1.10, +.36)
Declination-resid. SD	−0.242	20/24	96	stable	(−.98, +.45)
Adjective rate	+0.062	17/24	59	stable	(−.10, +.23)
Voice clarity (A)	+0.461	13/24	99	—	(−1.61, +2.80)
Voice clarity (B)	+0.315	13/24	99	[+.11, +.47]	(−1.51, +2.28)
Pitch variation (Hz)	−0.228	17/24	98	[−.37, −.16]	(−1.17, +.67)
Pause proportion	+0.196	14/24	98	sign flips	(−.79, +1.03)
Articulation (syll/s)	+0.108	17/24	96	stable	(−.59, +.86)
Sentiment polarity	+0.030	17/24	74	stable	(−.20, +.26)

difference can be present across the whole corpus, statistically significant within debates, visible in 20 or 21 debates out of 24, and pass every check in Section IV-E, and still tell us nothing about the twenty-fifth debate. One objection deserves an answer: perhaps 24 debates simply cannot produce a prediction interval that excludes zero. Adjective rate answers it: its between-debate variation is low ($I^2 = 59\%$) and its interval, $(−0.10, +0.23)$, is within a factor of two of clearing. The obstacle is inconsistency between debates, not sample size. Between 59% and 99.5% of the variation in these measures is between debates rather than between candidates.

E. Exploratory Same-Candidate Analysis

Only two candidates were observed both winning and losing a general election. We fit the discriminant with each entirely removed, then project their debates into the space. For George H. W. Bush the separation is complete: every debate he won scores above every debate he lost, with the model never having seen him (6/6; gap +0.173). For Donald Trump the gap is +0.005, essentially nothing (4/6). One speaker's winning and losing debates are also four years apart, which the standardisation controls for period but not for anything else that changed in the interim. Two cases cannot settle the exploratory question posed in Section I-D; we report them because they are the only direct evidence the corpus contains. Full debate-by-debate scores are in Appendix H.

TABLE XI

THE LANGUAGE MODEL’S DEBATE-LEVEL ACCURACY (MAJORITY VOTE OVER 3 RE-RANDOMISED REPEATS) BESIDE HOW OFTEN A SEPARATE, QUESTION-FREE PROBE IDENTIFIES THE ACTUAL DEBATE FROM THE SAME INPUT.

Input	Outcome accuracy	Exact debate identified
Descriptors only	13/24	0/24
Transcript only	24/24	24/24

The textbook remedy for sentences nested inside speakers, a random effect per speaker, cannot be estimated here. Regressing the outcome label on speaker identity across the 51 candidate-by-debate cells gives $R^2 = 0.811$: **speaker identity explains 81% of whether a candidate won**. Twelve of fourteen speakers appear in one outcome class only, so for them the two variables are the same variable; the variance of the outcome coefficient is inflated 5.3-fold and only 36 residual degrees of freedom remain. A speaker random effect would not make the analysis better powered here; it would make it better calibrated, and the honest output of a correctly calibrated model on this corpus is a very wide interval.

VIII. LLM JUDGMENT: ACCURACY, MEMORIZATION, AND GROUNDING

This section answers RQ4. Section IV-G gives the protocol; this section gives the result. It repeats Section VII’s diagnosis, arrived at independently, in a completely different kind of model.

A. Transcript-Only and Descriptor-Only Judgments

Given the anonymised transcript, the model calls the winner in 24 of 24 debates. Given only the descriptor table, with no text, it calls 13 of 24, a coin flip ($p = 0.340$). Read alone, the first number looks like the model successfully reasoning over communicative style from text. Section VIII-B shows that it is not.

B. Contamination Probe

We give the same model the same anonymised transcript with no prediction question attached, and ask it to name the debate. It names the specific real debate—year and both candidates—in 24 of 24 cases (Table XI). Given the descriptor table alone, it does so in 0 of 24. Presidential debate transcripts are some of the most quoted text in existence; stripping names and dates removes the labels but not lines like “I did not have sexual relations with that woman” or a candidate saying “you’re likable enough, Hillary.” **The language model’s transcript-based accuracy cannot be interpreted as evidence of generalisable political-language reasoning, because the model identifies the exact debate from the same anonymised input in every case.** This is the same failure mode Section VII-C found in the decision forest and the discriminant, arrived at by a completely different method: what looks like the modality that helps is the modality that leaks identity.

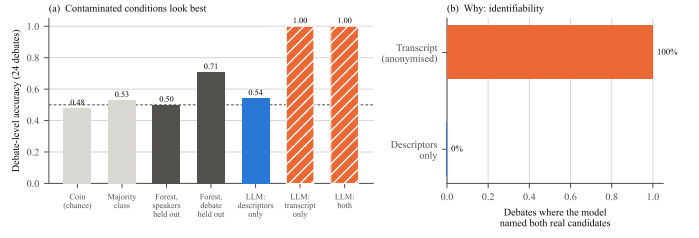


Fig. 7. (a) Debate-level accuracy of the language model, against the decision forest and the trivial baselines. The transcript condition (hatched) beats everything else in the paper; (b) shows why—the same model, asked only to identify the debate rather than predict its outcome, does so perfectly whenever it can see the transcript and never when it can only see numbers.

Figure 7 places the two results side by side. This is also the direct answer to a concern raised in Section I-C and revisited in Section X-E: whether a language model can identify a debate and its outcome from anonymised material is a number to report rather than a caveat to mention, and here it is 100% for text and 0% for numbers.

C. Grounding of Explanations

Every prediction call also cites, from a closed vocabulary, which evidence it relied on, which lets us check the citations by machine against the real data instead of taking them on faith. Given descriptors and citing a specific measure, the citation points the right way 75.8% of the time (303/400 citations); with the transcript added, 84.3% (348/413). Read alone, the second number looks like better reasoning with more evidence. Read alongside the contamination result, a less flattering explanation fits at least as well: once the transcript has told the model who wins, a numeric citation only has to sound consistent with an answer already fixed, not actually justify it. In the descriptors-only condition, where no such shortcut exists, the model’s most-cited measures are loudness, sentence length, voice clarity, grade level and speech rate, and its explanations read as plausible performance narratives—“Candidate B shows higher pitch variability, faster articulation rate, and more syntactically complex speech... which suggests a more dynamic and sophisticated delivery”—for a candidate who, in the 1988 debate that produced this call, went on to lose. Every one of those measures is one Section VII-B or Appendix F already found uncorrelated or confounded with winning. A fluent explanation is not the same thing as a correct one. We scan the transcript-only explanations for invented acoustic numbers, a figure the model was never given, and find none in any of the 72 transcript-condition responses; on this specific, narrow check the model does not fabricate.

D. Implications for LLM-as-Judge Research

Evaluating a language model’s judgment of famous political material needs at least two controls beyond accuracy: an identification or contamination probe, asked with no task question attached, and a check that cited reasons are actually consistent with the evidence presented rather than merely fluent. Neither control is specific to debates or to this model, and both are cheap to run. Section IX-F returns to what they generalise to.

IX. DISCUSSION

A. What Differs Between Eventual Winners and Non-Winners

Across four levels and 53 properties, we find no robust broad linguistic, semantic or discourse-level separation between the two groups (Sections V-A–V-B). We find one coherent pitch-control family, and within it one measure, declination-residual standard deviation, that is the strongest measurement-valid result in the paper (Section VI). The two original broad-screen pitch measures and this one describe the same underlying phenomenon at increasing precision: less overall range, less range relative to habitual pitch, and, most specifically, less local movement once each sentence’s own trend is removed.

B. Interpretation of Residual Pitch Stability

This analysis establishes a correlation between electoral outcome and one specific property of a pitch contour. It does not establish why that property might matter to a listener, and we ran no perception experiment to find out. What follows is a plausible account, built from findings already cited in this paper, and we offer it as interpretation rather than result.

Two production-side mechanisms and one perception-side mechanism point the same way. On the production side, planning an utterance in advance and executing it as one coherent unit is understood to produce a single, matched intonational gesture; formulating a sentence while speaking it, by reformulating a claim mid-sentence, hedging, or searching for the next word, is understood to introduce small disruptions to that gesture as the plan is revised in real time [6]. A rehearsed answer and an improvised one are not expected to look the same on this measure, and debate answers are frequently at least partly rehearsed in advance. Separately, fine control over the laryngeal muscles that set pitch is understood to be sensitive to physiological effort and strain: steadier voluntary control produces a cleaner pitch trajectory, and reduced control introduces the kind of small involuntary fluctuation this measure is built to detect [4]. On the perception side, the same voice-quality literature treats predictable, steady prosody as a standing cue to fluency and confidence, largely independent of what is being said [4]. If any of these mechanisms is operating here, a listener would not need to consciously notice the pitch contour to be influenced by it.

This interpretation connects directly to the factorial design we propose in Section X-E. That design pairs independently manipulated text and prosody with listener ratings; a declination-residual measure computed on the resynthesised audio in that design would test directly whether steadier residual pitch movement raises a listener’s rating of vocal control, holding the words fixed. Nothing in the present corpus can adjudicate between the mechanisms above, or confirm that any one of them is operating rather than some fourth explanation. The corpus establishes only that the acoustic pattern exists and is not an artefact of speaker identity, sentence length, or transcription treatment.

C. Speaker Identity Versus “Winning Style”

Twenty-four thousand candidate sentences do not equal 24,193 independent outcome observations: they come from fourteen people, and any generalisation claim has to hold up over those fourteen people, not over the sentence count. Holding out a debate while leaving its speakers’ other debates in training (Section VII-C) is not a held-out test of speaker identity. The gap between the two holdout schemes—19/24 to 10/24 for the discriminant, 17/24 to 12/24 for the forest, 17/24 to 11/24 for the weighted sum—is the size of that confound made visible. Feature stability under permutation importance (Table VII) does not imply transferable prediction: the same top-ranked measures survive speaker holdout, but the forest’s ability to use them does not.

D. What the Null Results Contribute

The null findings sharpen rather than weaken the positive one. Not sentiment. Not pronoun use, at any grammatical person. Not general syntactic complexity. Not pitch velocity, acceleration, or how often the pitch contour changes direction. Not most rhetorical categories, once we check the detectors that claimed to measure them. What remains after this many specific nulls is not “pitch matters, vaguely.” It is one measurable property of one signal, characterised precisely enough that a null result on a neighbouring measure (turning-point rate; kurtosis) rules out specific alternative explanations rather than merely failing to reach significance.

E. Implications for Multimodal Modelling

This lesson extends past presidential debates. A multimodal system trained on a small, repeatedly-observed population of subjects can improve its apparent performance by exploiting identity-bearing information, acoustic or textual, rather than the task-relevant behaviour it was meant to capture, and debate-level holdout alone will not reveal this if the same individuals recur across the held-out and training partitions. Three structurally different models (Section VII) reach this failure mode independently; a fourth, a large language model (Section VIII), reaches the text-modality version of the same failure by a completely different route.

F. Implications for LLM Research

Memorisation of famous transcripts is not a remote risk for language-model evaluation on any well-known text. It is directly measurable with a question-free identification probe, and here it was total for the raw transcript and absent for numeric descriptors of the same material. Partial anonymisation (names, party and dates removed, content untouched) was sufficient to demonstrate the contamination without destroying the communicative content the transcript condition existed to test, but it was not sufficient to prevent the contamination itself, and that is the more important number to report. A grounding requirement—citations checkable against real data, from a closed vocabulary—separates answer accuracy from valid reasoning in a way free-text explanation cannot, and we treat it as a minimum bar for LLM-as-judge use on any

material a model may have seen before, not only on debate transcripts.

G. Relationship to Previous Research

Rosenberg and Hirschberg’s [7] expectation that pitch and pausing would matter most is broadly supported: the only broad-screen measures that survive are pitch measures, and the pausing result, though it does not survive the speaker check, is large enough to have looked like a genuine finding without it. Künzel’s [10] warning about average pitch is directly borne out by the paper’s largest and least trustworthy effect (voice clarity, Section V-C). What this design adds, that a single-cycle or single-debate study could not, is the speaker-aware evaluation that turns “pitch range differs” into “pitch range differs, and specifically the residual component of it, in a way that does not yet predict an unseen debate.” This is a more qualified conclusion than the charisma literature’s framing, and we take it to be a more useful one.

X. LIMITATIONS AND FUTURE WORK

A. Corpus Limitations

- **Fourteen speakers is few.** Nine elections and fourteen people can reveal only differences that are nearly universal. A null result means no evidence, not evidence of nothing.
- **Third-party candidates are not like major-party losers.** Ross Perot contributes 1,024 sentences. Dropping him and recomputing is the only evidence we have on how much he matters.
- **One language, one broadcasting tradition.**
- **Two 2024 debates are excluded** for unverifiable speaker attribution (Appendix I). Including them changes several secondary results and none of the primary ones.

B. Measurement Limitations

- **Some timings are interpolated,** and missing acoustic data is uneven across alignment treatments (2.8% against 16.5%).
- **Interruptions are not represented.** No two candidate sentences overlap in time, including in the 2020 debate, which was defined by people talking over each other.
- **Sentence boundaries vary across the corpus,** and every per-sentence count depends on them.
- **A quarter of sentences are too short** for reliable voice measurement (Table XIV).

C. Inferential Limitations

- **The election result is not a measure of good speaking.** It is a shared external target, and no claim here depends on winning and speaking well being the same thing.
- **This is correlation.** Winners differ from losers in incumbency, money, party position and prior office. Incumbency can be added within debates at no cost and should be the first control a follow-up adds.

- **Every prediction interval reported includes zero.** An association can be real, significant, and repeatedly observed across this corpus without yet supporting a prediction about a debate we have not seen (Section VII-D).

D. LLM Limitations

- **One model, one API version, one evaluation date.** Every language-model result in this paper uses `claude-opus-4-8` on 2026-08-06 (Appendix G). We tested one prompting variation (chain-of-thought against direct) and it made no difference, but we cannot rule out that a different model or prompt would.
- **The training-data concern is measured, not assumed.** The contamination probe (Section VIII-B) shows the model identifies the exact debate, year and both candidates from an anonymised transcript in 24 of 24 cases. This corpus cannot be absent from the training data, its memorisation is directly measurable, and any result touching the transcript text has to be read through that fact.
- **Discourse-tag agreement here is instrument-versus-instrument, not instrument-versus-ground-truth.** We compare rule-based and model-based labels against each other (Section V-B); neither is validated against adjudicated human judgment. The chance-level agreement on rhetorical categories is informative about the instruments’ disagreement with each other, not proof that either is wrong.

E. Future Work

Two studies would extend this design most directly.

An expanded observational corpus. The limit on the substantive question is the number of elections, not the amount of speech. Adding vice-presidential, primary, senatorial and gubernatorial debates would keep the format while multiplying the speakers and the contested outcomes, and would supply many more people who both win and lose. Every check reported here carries over unchanged.

A controlled text×prosody perception experiment. In observed speech, what a candidate says and how they say it always arrive together from the same person, which is why no corpus of this kind can separate them. We propose the following design instead. Take an utterance; rewrite the words with a language model to be more or less charismatic, holding content fixed; resynthesise the delivery, holding words fixed; cross the two so each utterance appears in four versions; and have listeners rate vocal control, confidence, charisma, credibility and persuasiveness:

$$C = \beta_0 + \beta_1(\text{text}) + \beta_2(\text{prosody}) + \beta_3(\text{text} \times \text{prosody}) + u_{\text{speaker}} + u_{\text{listener}} + \varepsilon. \quad (4)$$

Now β_1 and β_2 are separately identified, because the experimenter assigned them rather than the candidate, and both random effects are estimable, because every speaker and listener appears in every condition, a property this corpus

lacks. Language models can rewrite text while holding content roughly fixed, and current speech synthesis allows independent control of pitch, energy, rate and style, which makes this design practical now.

Two cautions carry over from this paper, and the first is no longer hypothetical. Section VIII-B’s contamination probe confirms directly that a language model given an anonymised transcript from this corpus identifies the debate in every case; the same check has to run on whatever text a language model rewrites for the design above, and on the rewritten output itself, before any listener sees it, not only on the original transcript. Numeric descriptors did not carry the same signal in our check, but rewritten, resynthesised speech is a different object again and should not be assumed clean until tested. These are also fourteen of the most recognisable voices in America, so listeners will identify them; whether a listener recognised the speaker should be recorded and modelled, not assumed away.

XI. CONCLUSION

In this paper we describe a multimodal analysis of every televised U.S. general-election presidential debate from 1988 to 2020. We screen 43 linguistic, semantic, discourse and acoustic measures for a within-debate difference between eventual winning and non-winning candidates, and follow the one acoustic property that survives down to the level of the pitch contour itself. We find that eventual winning candidates show less residual pitch movement, once each sentence’s own broad rise or fall is removed, than eventual non-winning candidates: effect -0.242 standard deviations, present in 20 of 24 debates, stable across fourteen leave-one-speaker-out refits, uncoupled from habitual pitch, and, unlike a measurement artefact, stronger rather than weaker in longer sentences. Of 53 properties measured at four levels, this is the one pattern that survives every check we ran.

We demonstrate, however, that this association does not yet support reliable prediction for an unseen speaker or a new debate: its 95% prediction interval includes zero, for a structural reason rather than a statistical one. Only fourteen people have ever taken this stage, twelve of them once, and speaker identity explains 81% of the outcome label those fourteen people carry.

We show that this same structural fact is why a decision forest, a linear discriminant, a weighted sum, and a large language model all appeared, at first pass, to predict the winner, and why all four stopped working once we changed the evaluation to remove speaker or transcript identity rather than merely the debate under test. We conclude that speaker-aware validation—holding out every debate a candidate ever appears in, not just the one under test—and, for a language model, a question-free contamination probe alongside a closed-vocabulary grounding check, are not optional refinements to this kind of analysis. They are the difference between a predictive result and an artefact of who, or what text, the model had already seen.

The advantage of the speaker-aware and contamination-aware protocol used throughout this paper is that it is cheap to apply and specific to no one model: the same two holdout

schemes and the same two language-model checks generalise to any multimodal study of a small, repeatedly-observed population of speakers. Limitations of the present corpus and directions for extending it are given in Section X.

ACKNOWLEDGMENTS

The author thanks the maintainers of the open-source speech and language tools on which this work depends.

APPENDIX A DEBATE INVENTORY

TABLE XII
EVERY DEBATE IN THE CORPUS: DATE, CANDIDATES, AND WHICH TICKET WON THE ELECTION THAT FOLLOWED. AUDIO AND OFFICIAL TRANSCRIPTS ARE CITED BY DEBATE IDENTIFIER IN THE DEPOSITED DATA STATEMENT (APPENDIX I).

ID	Date	Candidates	Winner
1001	1988-09-25	G. H. W. Bush vs. Dukakis	Bush
1003	1988-10-13	G. H. W. Bush vs. Dukakis	Bush
1004	1992-10-11	Clinton vs. Perot, G. H. W. Bush	Clinton
1006	1992-10-15	Clinton vs. Perot, G. H. W. Bush	Clinton
1007	1992-10-19	Clinton vs. G. H. W. Bush, Perot	Clinton
1008	1996-10-06	Clinton vs. Dole	Clinton
1010	1996-10-16	Clinton vs. Dole	Clinton
1011	2000-10-03	G. W. Bush vs. Gore	Bush
1013	2000-10-11	G. W. Bush vs. Gore	Bush
1014	2000-10-17	G. W. Bush vs. Gore	Bush
1015	2004-09-30	G. W. Bush vs. Kerry	Bush
1017	2004-10-08	G. W. Bush vs. Kerry	Bush
1018	2004-10-13	G. W. Bush vs. Kerry	Bush
1019	2008-09-26	Obama vs. McCain	Obama
1021	2008-10-07	Obama vs. McCain	Obama
1022	2008-10-15	Obama vs. McCain	Obama
1023	2012-10-03	Obama vs. Romney	Obama
1025	2012-10-16	Obama vs. Romney	Obama
1026	2012-10-22	Obama vs. Romney	Obama
1027	2016-09-26	Trump vs. H. Clinton	Trump
1029	2016-10-09	Trump vs. H. Clinton	Trump
1030	2016-10-19	Trump vs. H. Clinton	Trump
1031	2020-09-29	Biden vs. Trump	Biden
1033	2020-10-22	Biden vs. Trump	Biden

APPENDIX B ALIGNMENT AND PREPROCESSING

Audio was converted to single-channel PCM at 16 kHz. For matching, both the recognised text and the official transcript were reduced to lower-case alphanumeric tokens with punctuation removed, contractions unexpanded and numerals as written. The match rate is the share of official-transcript tokens falling inside a matching block, found with a longest-matching-block algorithm. Routing thresholds were $\geq 85\%$ and 60–85%.

Recognition used Whisper [17], which gives word-level timings. Matched tokens take those timings directly; unmatched runs are interpolated between the nearest matched words, split by character count.

Montreal Forced Aligner [18] was run with the English ARPA acoustic model and dictionary. Default beam settings did not finish on ninety-minute recordings, so production runs used wider beams under a watchdog. It converged on 20 of 24 debates and is kept as an independent check.

APPENDIX C COMPLETE FEATURE DEFINITIONS

Pitch, loudness and voice-quality measures come from Praat [19] through `parselmouth`. The 88-parameter set comes from openSMILE [20], [21], which also supplies the semitone pitch measures and the second voice-clarity implementation. Parsing and tagging use spaCy [22] with `en_core_web_sm`, the smallest available model, whose weakest outputs are exactly parse depth and dependency length. Sentiment uses VADER [23], topic categories use Empath [24], and sentence vectors use Sentence-BERT [25].

A silent pause is a stretch of at least 150 ms during which loudness stays at least 25 dB below the ninetieth percentile for that sentence. The usual threshold after Goldman-Eisler [5] is 250 ms. The detector cannot see filled pauses, and whether those were transcribed at all varies across this corpus.

Table XIII names the 34 standardised measures used in Sections VII-A and IV-G, which overlap heavily with the full 43-measure broad screen (Section IV-B); the remaining measures are named individually in the results prose of Section V (jitter, shimmer, four readability scores, positive/negative sentiment, and the 35 topic categories treated as one diagnostic family rather than as individual screened measures).

TABLE XIII
THE 34 STANDARDISED MEASURES USED IN THE COMBINED AND LLM ANALYSES, BY FAMILY.

Acoustic (Praat)	Acoustic (openSMILE)	Linguistic
Mean pitch	Syllable rate	Clauses
Pitch SD (Hz)	Articulation rate (syll/s)	Subordinate clauses
Pitch range (Hz)	Semitone pitch SD/mean	Parse depth
Loudness	Semitone 20–80 range	Dependency length
Voice clarity (Praat)	Voice clarity (openSMILE)	Words per sentence
Pause duration		Word length
Pause count		Function words
Mean pause length		1st/2nd/3rd-person pronouns
Speech rate		Nouns, verbs, adjectives, adverbs
Articulation rate (w/s)		Reading ease, grade level
		Sentiment, positive, negative

For the frame-level pitch-contour measures (Section IV-C), the low-level-descriptor output of openSMILE (`F0semitoneFrom27.5Hz_sma3nz, 10 ms hop`) was run independently per sentence clip rather than sliced from a continuously smoothed whole-debate stream, which was tried first and rejected: continuous smoothing across sentence boundaries shifted the resulting statistics measurably at clip edges. Distributional measures use all voiced frames regardless of order; dynamic measures use only differences between temporally consecutive voiced frames; declination slope and its residual standard deviation come from an ordinary-least-squares fit of the semitone contour against frame time.

Four measurement problems are recorded for anyone reusing the corpus. *One pitch range for everyone*: the analysis range was 75–500 Hz for a corpus where average pitch runs from 102.5 to 195.9 Hz, and the fixed 75 Hz floor is the likely cause of the pitch-dependence of voice clarity. *Mismatched*

floors: Praat’s loudness default assumes a minimum pitch of 100 Hz against the 75 Hz used for pitch. *Voiced fraction* runs from 0.583 to 0.742 across speakers and correlates with voice clarity at +0.674 at speaker level, persisting within a single speaker in a single debate at a median of +0.327. *Reliability depends steeply on sentence length* (Table XIV), which is why the length check in Section V-C had to be run first.

TABLE XIV
VOICE-CLARITY RELIABILITY BY SENTENCE LENGTH. SPLIT-HALF RELIABILITY IS ODD-VERSUS-EVEN SENTENCES WITHIN ONE SPEAKER IN ONE DEBATE.

Sentence length	Mean clarity	SD	Split-half
0.3–1.0 s	8.79 dB	3.10	0.126
1.0–2.5 s	9.51 dB	2.42	0.476
2.5–6.0 s	9.96 dB	2.16	0.655
6.0 s+	10.30 dB	2.00	0.729
Candidate sentences: median 3.02 s, IQR (1.53, 5.71); 24.2% under 1.5 s, 13.1% under 1.0 s, 4.6% under 0.5 s.			

APPENDIX D FULL STATISTICAL RESULTS

Debate effects are removed by subtracting each debate’s mean. Cluster-robust variances use a sandwich estimator with a small-sample correction, referred to a t distribution with $G-1$ degrees of freedom, $G = 24$ for debate clustering and 14 for speaker clustering. With this few clusters the p -values should be read as generous rather than exact.

The randomization test enumerates all $2^{24} = 16,777,216$ ways of flipping the winner label within debates. Random-effects pooling follows DerSimonian–Laird, with the prediction interval computed as the pooled estimate $\pm t_{22} \sqrt{\tau^2 + SE^2}$. Leave-one-speaker-out recomputes the within-debate effect with one speaker’s sentences removed.

Several of the 43 measures overlap: 13 pairs correlate above $|r| = 0.7$, and the effective number of independent measures is 30.0 by the Li–Ji method. Correcting at $m = 30$ instead of $m = 43$ changes nothing.

The area under the curve (AUC) reported for the discriminant has a direct reading: take one sentence from a winning candidate and one from a non-winning candidate at random, and AUC is the probability the model scores the winning one higher. A model that has learned nothing scores 0.5; one that always gets the pair right scores 1.0. Unlike accuracy it does not depend on group sizes, which matters because the groups are unequal and because 12 of the 14 speaker folds contain only one group. 0.5 is the right baseline only when the held-out items are independent (Section VII-C).

The deposited version of this appendix carries the full table over all 43 measures, the treatment-split table, and all 35 topic categories.

APPENDIX E DISCOURSE ANNOTATION

The rule-based labeller assigns dialogue acts and rhetorical categories from sentence type, word and pattern matches,

question syntax, and first-person stance markers. Two of its categories are copies of the sentence-type column rather than independent detectors, and it returns zero for 13 codes. Table XV shows what fraction of each category’s hits come from a single string.

TABLE XV
WHAT THE RULE-BASED DETECTORS ACTUALLY FIRE ON. “SOLE TRIGGER” IS THE SHARE OF HITS PRODUCED BY ONE STRING AND NOTHING ELSE. “REP.” IS THE COUNT AFTER FIXING THE RULE; “EFF.” THE DIFFERENCE AFTER FIXING.

Category	Hits	Sole trigger	Rep.	Eff.	dir.
Unity framing	396	“United States” 80.1%	69	−0.004	10/24
Patriotic appeal	1,235	“America” 88.0%	115	−0.014	13/24
Historical ref.	265	bare year 63.4%	165	−0.028	15/24
Fear appeal	540	substring 30.9%	373	+0.007	11/24
Emotional appeal	435	“children”+“families”	—	—	—
Agreement	237	“absolutely”/“exactly”	—	—	—
Rejection	178	initial “no” 93.8%	—	—	—
Statement	22,629	= declarative 100%	—	—	—
Imperative	697	= imperative form 100%	—	—	—

Fixing the four fixable rules—anchoring at word boundaries and requiring a real unity or patriotic phrase—cuts unity framing from 396 hits to 69 and patriotic appeal from 1,235 to 115. Between 94 and 97 per cent of what these categories counted was something else. What survives appears in 0.3 to 0.5% of sentences with differences no larger than 0.014.

The corpus was also labelled by a large language model through a batch API, using a prompt that assigns dialogue acts first and rhetorical categories second. The model returned a parseable answer for 100.0% of sentences, assigning at least one dialogue act to 99.86% and at least one rhetorical category to 71.05% of candidate sentences. Table XVI gives category-level agreement; the two labellings agree best on categories readable from the grammar (questions, thanks) and worst on the interpretive ones.

TABLE XVI
AGREEMENT BETWEEN THE RULE-BASED AND MODEL LABELS OVER 24,193 CANDIDATE SENTENCES. PSA TREATS NEITHER AS THE REFERENCE. PABAK AND AC1 CORRECT FOR HOW RARE EACH CATEGORY IS.

Category	Rule	Model	PSA	κ	PABAK	AC1
Thanks	182	217	0.852	0.851	0.995	0.998
Wh-question	212	376	0.599	0.594	0.980	0.990
Polar question	146	290	0.564	0.561	0.984	0.992
Explanation	1,478	869	0.415	0.387	0.886	0.937
Personal stance	1,424	4,865	0.389	0.327	0.682	0.795
Fear appeal	540	1,674	0.237	0.210	0.860	0.923
Historical ref.	265	1,343	0.208	0.193	0.895	0.944
Patriotic appeal	1,235	282	0.190	0.174	0.898	0.946
Imperative	697	1,234	0.177	0.146	0.869	0.929
Unity framing	396	1,099	0.060	0.037	0.884	0.938
Rebuttal	3	4,653	0.001	0.001	0.616	0.767
Informative stmt.	2,475	0	0.000	0.000	0.795	0.887

Whole-set agreement (Krippendorff α , MASD):
dialogue acts +0.199; rhetorical categories +0.012.

Chain-of-thought variant. A natural objection to the direct prompt above is that it never gives the model room to reason—it goes straight from sentence to codes. We re-ran the labelling on a stratified sample of 3,001 candidate sentences

(12.4% of the corpus, proportional across all 24 debates) with one change: the output schema requires a short free-text reasoning field, written before the `damsl` and `beads` codes, so the model must commit to an explanation before it commits to a label—the same chain-of-thought design Wei et al. [12] showed can improve accuracy on tasks with a checkable right answer. Same model, same codebook, same prompt otherwise (Appendix G). It does not help. Figure 8 gives Krippendorff’s α against the rule tagger for both prompting styles, recomputed on the identical sample: dialogue acts move from +0.167 direct to +0.169 with reasoning; rhetorical categories move from −0.030 to −0.002, both inside sampling noise of each other and both the same chance-level result as the full-corpus comparison above. Chain-of-thought and direct prompting agree with *each other* at $\alpha = +0.564$ for dialogue acts and +0.536 for rhetorical categories—moderate agreement, well short of the near-unanimity two runs of a deterministic rule tagger would show—while both remain equally uncorrelated with the rule-based labels. Reasoning also makes the model slightly more conservative: it assigns a dialogue act to 98.20% of the sample and a rhetorical category to 66.61%, against 99.83% and 70.68% for direct prompting on the identical sample. The instrument’s problem was never that the model failed to explain itself; it is that the categories it is asked to detect are not observable from either the grammar or an explanation, and asking for one does not manufacture the signal.

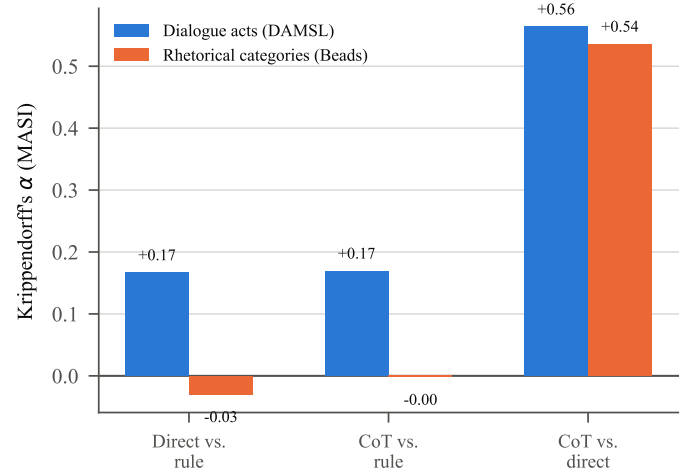


Fig. 8. Chain-of-thought prompting does not close the gap with the rule-based tagger. Both prompting styles land at the same chance-level agreement with the rules; what changes is how much the two LLM conditions agree with each other.

APPENDIX F SUPPLEMENTARY MULTIMODAL MODELS

Weighted sum. Each measure gets a weight equal to how well it separates the groups in training debates T :

$$w_j = \frac{1}{|W_T|} \sum_{i \in W_T} z_{ij} - \frac{1}{|L_T|} \sum_{i \in L_T} z_{ij}, \quad (5)$$

weight vector scaled to unit length, each sentence scored as $S_i = \sum_{j=1}^{34} w_j z_{ij}$, weights learned on 23 debates and scored on the twenty-fourth, rotating through all of them.

TABLE XVII
COMBINING MEASURES, AGAINST THE TWO MEASURES THAT SURVIVED
ON THEIR OWN.

	Single measures		Composite (34 measures)	
	Semitone 20–80 range	Semitone SD/mean	hold out debate	hold out speakers
Mean per-debate effect	−0.243	−0.370	+0.390	+0.070
SD across debates	0.289	0.358	0.603	0.519
Debates in direction	21/24	20/24	17/24	11/24
Sign-test p	0.0003	0.0015	0.064	0.84

On held-out debates the weighted composite reaches +0.390 standard deviations, a 60% improvement on the semitone percentile range alone, but markedly less consistent (SD across debates 0.603 against 0.289) and, under speaker holdout, falls to +0.070—a single measure needing no training is more reliable than a weighted combination of 34.

Discriminant loadings. Averaged to the 51 speaker-by-debate cells, the discriminant separates winning from non-winning cells at an area under the curve of 0.907. Table XVIII reports the correlation between each measure and the discriminant score, more stable than raw weights on correlated measures.

TABLE XVIII
WHAT THE DISCRIMINANT LEANS ON. POSITIVE MEANS HIGHER AMONG
WINNING CANDIDATES. EVERY MEASURE IN THE TOP TEN IS ACOUSTIC.

Measure	Loading
Voice clarity (openSMILE)	+0.628
Semitone pitch SD/mean	−0.587
Voice clarity (Praat)	+0.462
Pitch SD (Hz)	−0.422
Semitone 20–80 pitch range	−0.410
Pitch range (Hz)	−0.317
Articulation rate (syll/s)	+0.235
Mean pitch	+0.189
Pause duration	+0.186
Articulation rate (words/s)	+0.181

Every measure in the top ten is acoustic; nothing syntactic, lexical or semantic appears until well below. The highest-loading measure is the one shown in Section V-C to track the speaker’s own pitch at $r = +0.78$ to $+0.90$. Consistently with that, 81.0% of the variance in the cell scores is explained by which speaker the cell belongs to.

Principal component analysis. PCA asks whether the outcome happens to line up with the largest-variance directions among the 34 measures without using the labels at all.

PC1 (20.9% of variance) is a sentence-length dimension separating the groups at +0.006 standard deviations—nothing. PC2 is speaking rate. The component that best separates the groups is PC5, at +0.359, accounting for only 6.4% of the variance. A hand-built weighted sum, a supervised discriminant, and an unsupervised decomposition that ignores the labels entirely all reach the same place: what structure exists in these 34 measures is mostly about who is speaking.

LLM “both” condition. Given the transcript and the descriptor table together, the model calls 24 of 24 debates correctly, agreeing with the transcript-only condition (Section VIII-A) on all 24 debates—adding numeric descriptors to a transcript the

TABLE XIX
PRINCIPAL COMPONENTS OF THE 34 STANDARDISED MEASURES.

	% of variance	Separation of groups	Dominant measures
PC1	20.9	+0.006	Sentence length, grade level
PC2	10.6	+0.136	Articulation rate, speech rate
PC3	7.9	+0.203	Pitch SD, reading ease
PC4	6.8	+0.070	Mean pitch, voice clarity
PC5	6.4	+0.359	Voice clarity (both tools)
PC6	5.9	+0.044	Sentiment

12 components needed for 80% of variance.
At cell level, speaker explains 83.6% of PC1 and
86.6% of PC2 positions.

model can already place in history changes nothing, because the transcript had already settled the answer.

APPENDIX G LLM METHODS AND REPRODUCIBILITY

This appendix specifies every language-model call in the paper: the original discourse tagging and its chain-of-thought variant (Appendix E), and the cross-modal winner prediction with its contamination probe (Sections IV-G, VIII). The goal is that another researcher could reissue every call from this description alone.

Model. All components use the same model, `claude-opus-4-8`, accessed through Anthropic’s Messages API. Anthropic does not publish a parameter count for this model, so none is reported here; reporting one would imply a precision we do not have. Every call uses structured (`json_schema`) output: the model’s response is constrained at generation time to match a schema we specify, rather than parsed after the fact from free text. All calls were issued on 2026-08-06.

Sampling. This model deprecates the `temperature` parameter—the API rejects a request that sets it. Repeated calls therefore draw on whatever sampling variability the API applies by default, not on a caller-chosen temperature; where the paper reports a repeat count, that is the only lever available for measuring stability.

Discourse tagging. System prompt, codebook and full JSON schema are reproduced verbatim in `Sahana Project/tagging_prompt.md` (direct) and `Sahana Project/cot_tagging_prompt.md` (chain-of-thought). Both send 10 sentences per request, batched for throughput, with the system prompt and codebook cached (`cache_control: ephemeral`) across requests. The only schema difference is one field, `reasoning`, which the chain-of-thought variant requires before `dams1` and `beads`; structured output fills schema fields in the order they are declared, so this forces the free-text reasoning to be generated before the codes rather than after them.

Cross-modal winner prediction. System prompt, both JSON schemas (descriptor-citing and theme-citing), the contamination-probe prompt and schema, and the closed evidence vocabularies are reproduced verbatim in `Sahana Project/crossmodal_prompt.md`. One call is issued per debate, per condition, per repeat, with a freshly randomised

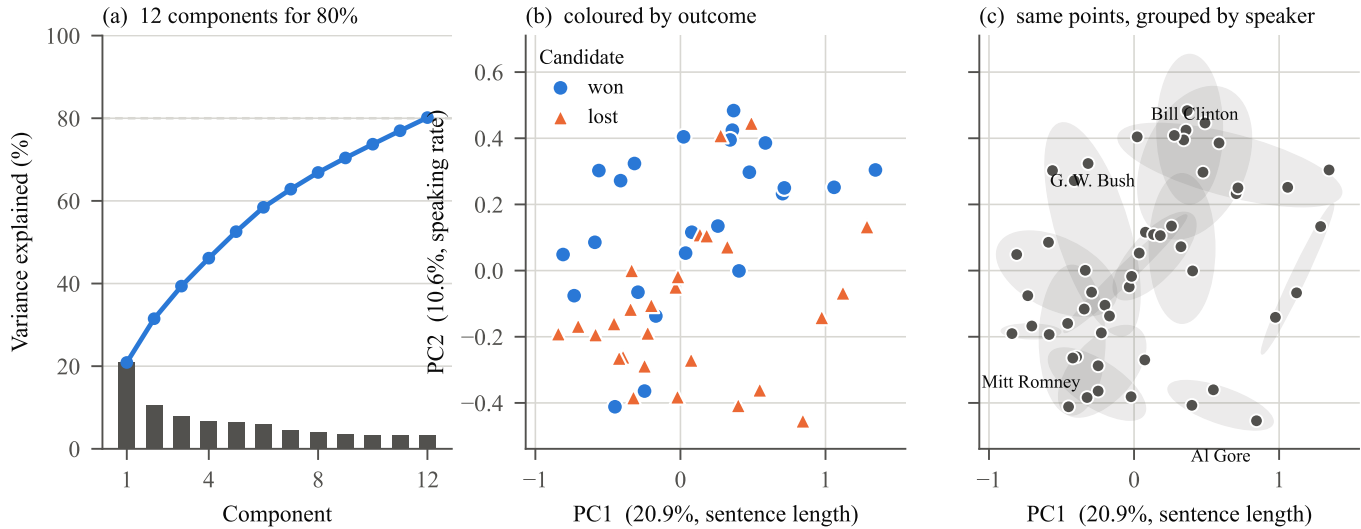


Fig. 9. Principal component analysis of the 34 standardised measures. (a) Twelve components are needed for 80% of the variance. (b) Each point is one candidate in one debate, coloured by outcome. (c) The same points, with an ellipse around each speaker’s debates: apparent separation by outcome in (b) coincides with the speaker grouping in (c).

Candidate-A/B/C letter assignment on each repeat, shared across conditions within a repeat so predictions are comparable trial-for-trial. The contamination probe issues one further call per debate per input type, with no repeats, since it is a single-shot identification question rather than a graded judgement. Every call uses a fresh, empty context, so agreement or disagreement between conditions reflects the input alone.

Anonymisation. Candidate names, party affiliation and the debate date are removed from every input; moderator turns are dropped entirely, leaving only candidate sentences in original chronological order. This is a partial defence at best: a transcript’s content—quoted lines, named policies, the other candidate’s real name spoken aloud mid-turn—was not scrubbed, because doing so would have removed the communicative content the transcript condition exists to test. The contamination probe measures what leaks through this partial anonymisation rather than assuming it is complete; 24 of 24 transcripts were identified regardless of the redaction.

Grounding check. A cited measure counts as consistent with a debate’s real data when the candidate it names as favoured actually has the higher within-debate standardised value for that measure (a margin of less than 0.15 standard deviations between the two candidates is scored as “similar” and excluded rather than forced to a side). This is an automatic, closed-vocabulary check, not a human judgement of whether the reasoning sounds plausible.

APPENDIX H EXPLORATORY SAME-CANDIDATE ANALYSIS

APPENDIX I 2024 EXCLUSION AND KNOWN CORPUS ISSUES

Two 2024 debates exist in the corpus but are held out, because we could not verify who was speaking. Their official transcripts are paraphrased news summaries sharing only 5.2% and 5.5% of their words with the audio, so the recognised text was used as the transcript and speakers were assigned by

TABLE XX
WHERE THE SAME SPEAKER LANDS WHEN THEY WON AND WHEN THEY LOST. THE DISCRIMINANT WAS FITTED *without* THAT SPEAKER, THEN THEIR DEBATES PROJECTED IN. SCORES ARE RELATIVE TO THE OPPONENT ON THE NIGHT.

Speaker	Debate	Outcome	Score
G. H. W. Bush	1988, first	won	−0.021
	1988, second	won	−0.066
	1992, first	lost	−0.224
	1992, second	lost	−0.203
	1992, third	lost	−0.223
	gap (won − lost) winning above losing		+0.173 6/6
Donald Trump	2016, first	won	+0.010
	2016, second	won	−0.018
	2016, third	won	−0.025
	2020, first	lost	−0.034
	2020, second	lost	+0.002
	gap (won − lost) winning above losing		+0.005 4/6

diarization [26]. That diarization was never checked against hand-labelled data and we have no error rate for it.

Four things prompted the exclusion. In the June 2024 debate one candidate is credited with 786 sentences and the other with 256—three to one, in a format with timed turns and muted microphones. In the September debate about 285 sentences sit in twelve or more leftover speaker clusters. Across the pair, 1,162 sentences are labelled *audience*, in two debates with no audience. And candidates account for 43.9% of sentences in these two, against 76 to 86% everywhere else.

The confirmation is what happens on removal: across the remaining 24 debates, unattributed sentences drop to zero and audience sentences to 287. All of the unattributed material and 80.2% of the audience material came from those two debates. Including the pair changes several results: noun rate flips sign, from −0.012 to +0.043; sentiment roughly halves;

the embedding axis rotates (cosine similarity 0.895 to the one reported here); and 19 of the 50 most extreme sentences at one end come from this 9.5% of the data. These transcripts also contain no truncation marks and no filled pauses at all, so interruption and self-correction are missing by construction.

Availability

The corpus is deposited with a permanent identifier. Released material covers the annotation layers, the per-sentence features and the checking scripts. Alignment scripts point at audio sourced elsewhere, which cannot be redistributed. A data statement, a column dictionary and the annotation codebook accompany the deposit.

Known problems

- 25 candidate sentences of eight words or more are exact duplicates within a debate, 17 of them in the first 1988 debate. One pair has the same text attributed to two different candidates, so speaker attribution is not verified anywhere in the corpus.
- 8 candidate rows share an identical time span—one group of nine sentences assigned a zero-length window.
- 238 candidate rows have no duration, almost all in one debate, which is also where every extraction error occurs.
- Acoustic extraction fails on 4.5% of candidate rows: 3.2% too short, 0.9% errors, split 2.8% against 16.5% across the two alignment treatments.
- No two candidate sentences overlap in time, so interruptions are not represented.
- Two rule-based categories are copies of the sentence-type column, and 13 codes are always zero.

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